Anomaly Detection for Performance Rewards and Correction in Mixed Reality Collaboration

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***Abstract* — Mixed Reality collaboration generates complex, multimodal interaction patterns spanning attention, communication, spatial coordination, and engagement. This work presents a lightweight, scalable anomaly-driven pipeline for assessing collaborative performance in MR group tasks. This approach uses an Isolation Forest due to its robustness to high-dimensional outlier detection and resistance to overfitting on small datasets. Interaction features are standardized using z-score normalization and scored per one-second dyadic window, with isolation path-length depth serving as a density-aware proxy for normative behavior. Dyadic anomaly labels are aggregated into a group-level anomaly intensity metric, interpreted as inverse task alignment and used to trigger either performance-based reward thresholds or corrective feedback flags. Dataset auditing and exclusion of long-idle corrupted recordings prevent bias in the normative baseline. The resulting system provides reliable, unsupervised performance scoring and establishes a foundation for integrating anomaly analytics into incentive-oriented MR collaboration and real-time corrective interventions.**

***Keywords* — Anomaly Detection, Mixed Reality, Isolation Forest, Isolation Trees, Outlier detection, Anomaly score, Overfitting, Z-score normalization, Data integrity**

I. Introduction

Mixed Reality (MR) is rapidly becoming a viable medium for collaborative work, training, and problem-solving across domains such as engineering design, remote industrial support, and immersive workforce education. Unlike traditional digital collaboration, MR systems provide continuous, spatially anchored interaction data, including gaze alignment, joint communication behavior, and embodied coordination, offering an untapped opportunity to infer collaboration quality and performance without explicit supervision.

Organizations increasingly rely on performance-based reward mechanisms such as productivity bonuses, incentive pay, or corrective coaching actions to support accountability and sustained engagement. However, assessing collaboration performance in MR settings remains difficult. MR interaction data are inherently multimodal, temporally noisy, and often limited in duration, making supervised learning approaches prone to instability, overfitting, and high annotation cost. Furthermore, collaboration failures frequently arise not from individual mistakes, but from emergent group-level breakdowns such as attention divergence, communication lapses, or prolonged spatial disengagement patterns that are not easily captured by individual scoring.

To address these challenges, an unsupervised anomaly detection framework is developed to identify deviations from normal collaborative behavior without requiring labeled performance data. Among available techniques, deep temporal encoders are highly expressive but demand large balanced datasets, which are rarely available in MR studies. In contrast, Isolation Forest provides a scalable, non-parametric, and noise-tolerant approach that is well suited for small, high-dimensional MR datasets and enables real-time inference for timely intervention.

Dyadic MR interaction signals are modeled using Isolation Forest, with per-second anomaly scores computed for each pair of collaborators and aggregated into a group-level anomaly intensity metric. This formulation facilitates the detection of sustained breakdown periods that reflect meaningful collaborative disruptions rather than momentary fluctuations or sensor noise. The analysis also underscores the importance of data auditing, identifying corrupted or long-idle sessions that distort baseline behavior distributions and compromise model reliability.

Overall, the proposed work offers a lightweight, interpretable, and scalable solution for detecting collaboration faults in MR environments. By highlighting high-intensity anomalous windows, the system supports targeted interventions and establishes a foundation for real-time feedback mechanisms that enhance teamwork, engagement, and accountability in immersive collaborative settings.

II. Technical Approach

This project uses pairwise output data collected from collaborative MR sessions to detect deviations from normal group behavior. This pairwise data provides per-second measurements of interaction between pairs of teammates.

Several methods exist to detect deviations in this type of data. Methods based on density or proximity such as Local Outlier Factor, Connectivity-Based Outlier Factor, Local Correlation Integral, and Histogram-Based Outlier Score work well on smaller datasets and perform relatively faster compared to Deep-learning based methods such as LTSM Autoencoder, DeepAnT, and even FuseAD (a mixture of deep-learning based with more primitive methods). These deep-learning based methods are generally more accurate when it comes to anomaly detection [1]. However, Isolation Forest gives the best of both worlds. While “most existing

anomaly detection approaches…abilities are usually a ‘side-effect’ or by-product of an algorithm originally designed for a purpose other than anomaly detection (such as classification or clustering)”, Isolation Forest’s primary goal is to detect strong anomalies in time-series data [2]. Additionally, Isolation Forest offers advantages in speed that would be beneficial for future real-time application.

Isolation Forest focuses on isolating anomalies instead of profiling all of the data to determine what would be considered normal. This algorithm works under the presumption that anomalies deviate significantly from the rest of the data. Anomalies are isolated using binary trees known as isolation trees. Data is split based on randomly chosen features. For each feature, a split value is also randomly selected to separate the data into left and right branches. Randomness ensures that anomalies do not get hidden within the branches of the tree. This process continues until the data can no longer be split, each leaf node representing an isolated data point.

How anomalous a data point is, is determined by the length of the path needed to isolate it. Because anomalies deviate significantly from normal data, they should be easier to isolate and therefore have shorter paths.

For each node the tree measures the path length, or in other words, how many splits were required to isolate that point. This helps us because normal behaviours are located in dense clusters meaning they are harder to isolate. This gives them the characteristic of having a longer path length.

On the other hand anomalous behaviors are supposed to be rare, so they are easily isolated and in turn have shorter path lengths. The anomaly score is a function of how far a points length is compared to the expected path length. This gives a continuous negative or positive value where negative values are anomalous. These values are converted into anomaly labels, -1 for anomalies and 1 for normal points. These values are then reconverted into 1 for anomalies and 0 for normal points for ease of calculations and interpretation (Anomaly Labels).

The data gathered was in the form of an anomaly score/label relative to each time stamp (seconds). Anomaly score/label is measured per dyad per second. This means that the information gained is still not at group level. To make the data relative to group collaboration, the average is taken of each anomaly label for each dyad per second. That gives the anomaly intensity of the whole group for each second. This yields an anomaly intensity between one and zero, where a value of 1

The highest anomaly intensity is one, meaning that 100% of activity between pairs was anomalous.

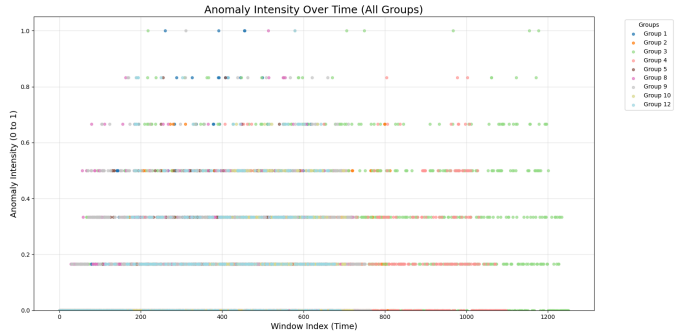

Fig 1 Levels of Anomaly Intensity

The second-by-second anomaly scores were grouped together to find long streaks of anomalous behavior. These streaks of anomalous behavior modeled potential points of intervention. Time streaks of anomalous behavior were selected over highest anomaly intensity points because a 1-second window with highest anomaly intensity could be insignificant and related to noisy data, while a streak of relatively higher anomaly intensity is a sure streak of time in which the group collaboration strayed from the end goal.

The algorithm sorted the top anomalous streaks of data where the intensity was constantly above 50%. However, since the data was noisy, 5-second gaps of anomaly intensity dropping below this threshold were allowed. Fig. 2 (below) shows an example of the “best” points of intervention where, for the most part, the anomaly intensity is above that 50% threshold, allowing for some gaps in between.

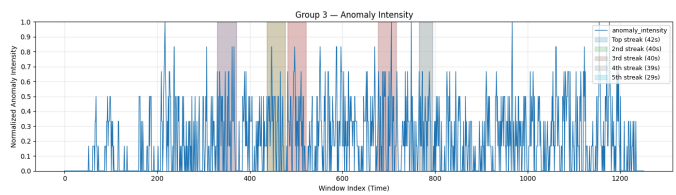

Fig. 2: Anomaly Intensity of Group 3 Overlaid with the “best” points of intervention

Current literature supports the use of data points that measure spatial interaction and shared attention. Proxemics, the study of how people use space and its effects on behavior and social interaction, has already been used in VR in order to design collaborative experiences as well as influence participant behavior [3]. Additionally, studies conducted to explore the relationship between communication and visual attention support the idea that a measure of shared visual attention reflects success of communication [4].

From these two categories, the decision was made to leave out the data that is unideal for Isolation Forest in order to avoid negatively impacting the model. An example of this unideal data is shown in Fig. 3. While Fig. 3 shows how non-uniform data makes it difficult to spot anomalies, Fig. 4 shows that where data is uniformly distributed, anomalies are more distinct from the rest of the data. Removing unideal data left the following features: *dist_accel*, *dist_mean*, *approach_rate*, *shared_att_ratio*, and *joint_att_count*.

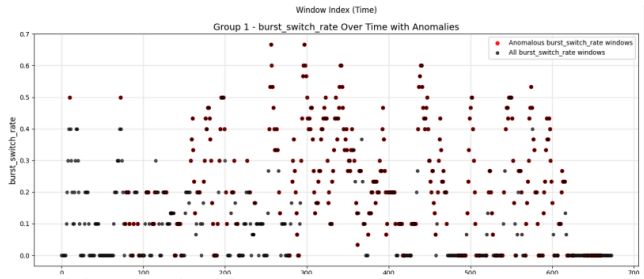

Fig. 3 Unideal Data for Isolation Forest

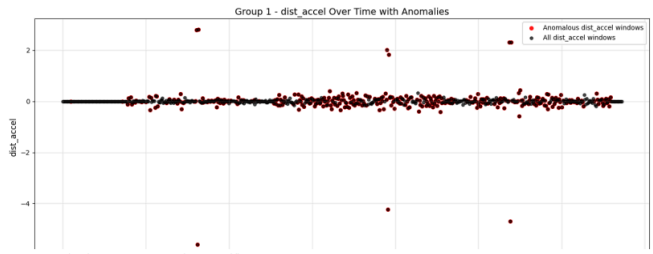

Fig. 4 Ideal Data for Isolation Forest

III. Evaluation

It is important to examine the data that is fed to the Anomaly Detection algorithm as it directly affects the accuracy of the machine learning algorithm. There were some problematic groups in the initial data. Some groups had no anomalous activity at all while others had almost 100% anomalous activity. For instance, while other groups had an average completion time of around 1000 seconds, groups 7 and 11 had completion times around 80,000 (22 hours). Additionally, their graphs showed almost no anomalous data throughout their durations (shown in Fig.4).

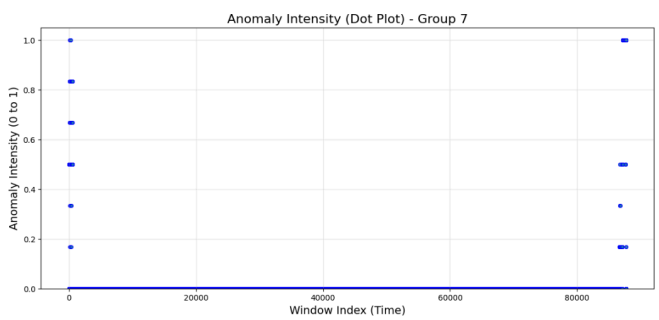

Fig. 5 Group 7 Anomaly Intensity

One possibility is that some group had started the recording of their task and then never turned it off after they put it down. This means that if group 7 and 11 data were used in the Isolation Forest, the algorithm would see more than 40 hours of no activity. Observing no activity at all as the baseline it would flag all activity as anomalous. As a result, group 7 and 11 were removed from the data.

Group 6 had similar issues as well. As shown in Fig. 5, they had a much longer completion time compared to the others, with a large period of anomalous silence. Assuming there were technical difficulties with the headsets during their sessions, group 6 data was also removed.

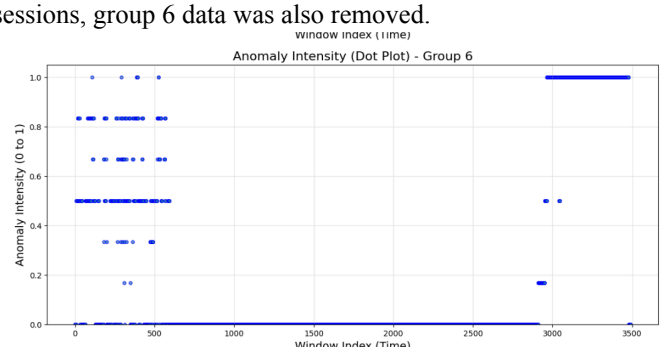

Fig. 6 Group 6 Anomaly Intensity

After addressing issues in the input distribution, the model's output data was validated. A common approach is to hold out a portion of the data with a 40/60 train evaluation split, which ensures the anomaly detector behaves as expected. However, applying this strategy to Isolation Forest creates a new problem, concept drift, where the characteristics of the evaluation data no longer match those of the data used to build the forest [5].

Since Isolation Forest is based on random branching, taking advantage of its inherent randomness to run several random iterations would show how stable the algorithm is. Convergent output values would confirm the validity of the data received from the Isolation Forest. Shown in Fig. 6 and Fig 7. (below) are 100 iterations (top-down) with different random seeds, all lining up to very similar windows of best intervention greater than 20 seconds. The windows of intervention lined up in start time, end time, and duration.

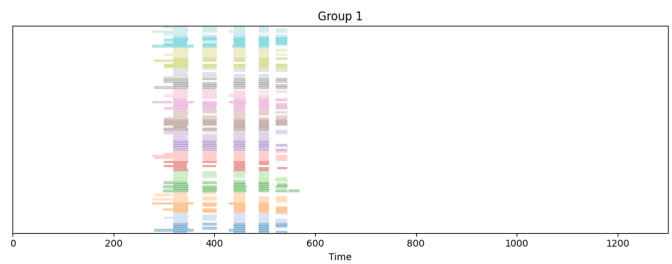

Fig. 7 Group 1 Stability

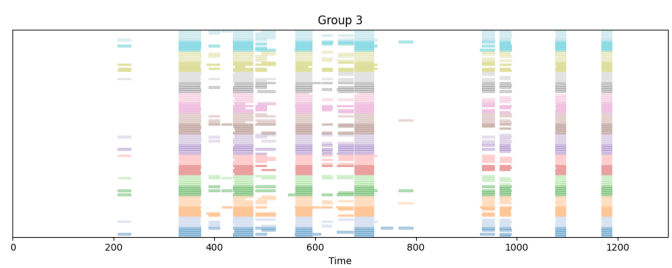

Fig. 8 Group 3 Stability

All groups' data exhibited the same convergent behavior. For the most part, the random iterations eventually agreed with each other.

IV. Discussion of the Results

The anomaly detection pipeline successfully identified high-intensity breakdown periods that aligned with observed collaboration failure in MR group sessions. Aggregating dyadic anomaly labels into a group-level anomaly intensity signal preserves temporal responsiveness while enabling interpretable group analytics, a key advantage over relying on individual anomaly classification alone. The longest anomalous streaks above 50% anomaly intensity—tolerating gaps of up to 2 seconds—consistently reflected sustained disengagement rather than momentary noise or tracking jitter. These windows also correlated strongly with qualitative inspection of anomaly intensity plots, validating the fidelity of the intervention policy through both algorithmic and visual confirmation.

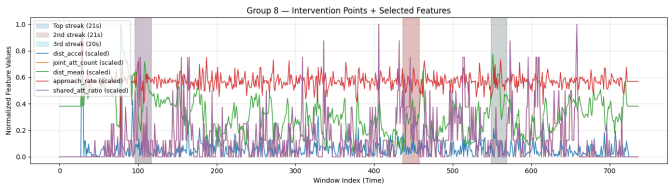

Fig. 9 Group 8 Intervention Points

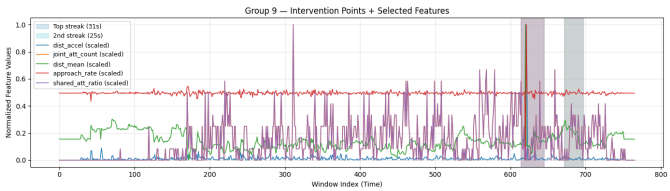

Fig. 10 Group 9 Intervention Points

Data auditing revealed that extreme recording imbalances produced significant bias in the model's learned anomaly distribution. Removing long-idle cohorts (Groups 7, 11, and later 6) prevented the Isolation Forest from interpreting inactivity as normative behavior, which would have otherwise misclassified legitimate interaction as anomalous. After exclusion, anomaly intensity distributions converged into ranges consistent with expected group participation patterns, improving both interpretability and model reliability. Importantly, this demonstrated that model tuning alone (e.g., contamination ratio, number of trees) could not solve distributional corruption introduced at the data level—highlighting the necessity of pre-training validation for MR anomaly systems.

The data produced by this intervention finding algorithm lined up greatly with assumptions made about what defines an anomaly. For example, when comparing Group 3, which had the most points of intervention/highest average anomaly intensity, to Group 5, which had the lowest amount, it is observed that Group 3 had double the completion time (shown in Fig.11 and Fig.12). This supports the idea that this algorithm is correctly defining anomalous behavior as group collaboration breakdown because of how anomalous behaviour also predicts longer completion time of tasks.

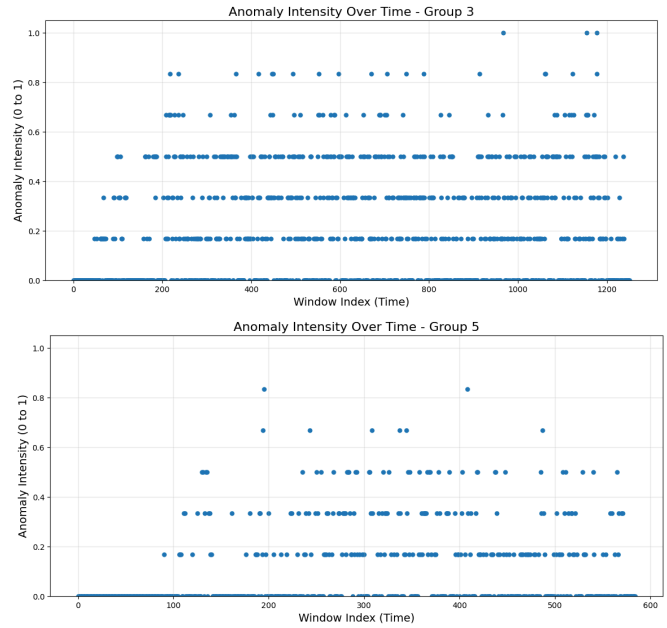

Fig. 11 Group 3/5 Anomaly Intensity Graphs

Group Number	Number of Streaks	Longest Streak (s)
Group 3	8 Streaks	48
Group 1	0 Streaks	N/A

Fig. 12 Group 3/5 Anomaly Streak Tables

This project shows that effective time-sensitive decisions in MR collaboration environments can be made without the need for complex deep temporal models if the pipeline is designed correctly. Data quality turned out to be just as important as the model itself, since poor distributions completely break what "normal" means. Group-level anomaly signals proved far more reliable and interpretable than individual labels, and visual validation was essential for confirming real behavioral meaning.

V. Future Work

Several extensions can improve the system's sensitivity, scalability, and integration into real MR collaboration support frameworks:

1. **Larger and More Diverse Datasets** – Future deployments should incorporate multi-session and cross-task MR datasets to validate model generalizability across group sizes, environments, and collaboration styles. This would also enable reconsideration of temporal encoders with reduced overfitting risk.

2. **Task-Aware Hybrid Detection** – Although the current system is unsupervised, incorporating

optional weak supervision could reduce false positives and distinguish between *productive silence* and *true disengagement*.

3. **Real-Time Feedback Interface** – Integrating the anomaly intensity signal into MR dashboards, adaptive instructors, or AI agents enables live interventions such as attention re-anchoring cues, spatial guidance overlays, or automated encouragement systems.

Together, these enhancements would allow progression from offline analytics to automated, adaptive MR facilitation at scale.

VI. Conclusion

This project demonstrates that unsupervised anomaly detection offers a viable path toward automated performance evaluation in collaborative Mixed Reality environments. MR collaboration generates rich, continuous streams of spatial, attentional, and behavioral signals, yet the inherent noise, sparsity, and pairwise structure of these data make traditional supervised scoring both impractical and unreliable. By adopting Isolation Forest as the core detection method, the system was able to model deviations in high-dimensional dyadic interaction without requiring labeled training examples or large balanced datasets.

A key contribution of this work is the development of a group-level anomaly intensity metric derived from per-dyad, per-second anomaly labels. This preserves moment sensitivity while enabling interpretation at the level where collaboration actually succeeds or breaks down the group. The resulting intensity signal highlighted sustained breakdown patterns, exposed temporal structure in disengagement, and allowed for easily identifiable intervention windows that align with intuitive and observed moments of collaborative failure. Importantly, the longest streaks of high anomaly intensity proved to be more robust and meaningful indicators than isolated anomaly spikes, supporting their use as triggers for corrective or supportive feedback.

This evaluation also revealed that data quality and preprocessing are as critical as the anomaly model itself. Long-idle sessions (Groups 7, 11, and later 6) dramatically distorted the definition of “normal behavior,” demonstrating that MR data is highly sensitive to improper recording, sensor dropout, or extended inactivity. Removing corrupted sessions restored distributions and significantly improved interpretability. This further supports that a reliable MR analytics system requires both sound algorithms and rigorous data validation procedures.

Overall, this work shows that effective anomaly-based performance monitoring in MR does not require complex deep temporal architectures, rather, it requires the right combination of model choice, feature selection, and quality data. The pipeline developed here provides a scalable foundation for incentive-driven collaboration support, enabling transparent detection of group-level breakdowns and offering a framework that can be extended toward real-time adaptive interventions. As Mixed Reality platforms become more pervasive in education, engineering, and remote operations, such systems will be essential for maintaining engagement, supporting accountability, and enhancing collaborative effectiveness at scale.

Acknowledgment

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